

```
In [1]: # Program 10: Independent two-sample t-test
import pandas as pd
from scipy import stats

FILE = r"D:\Datascience-MachineLearning\Datasets\titanic.csv" # <-- change to your file path
GROUP_COL = "Sex" # column that splits the data in two groups
VALUE_COL = "Age" # numeric column to compare

df = pd.read_csv(FILE)
g1_name, g2_name = df[GROUP_COL].dropna().unique()[2:]

group1 = df[df[GROUP_COL] == g1_name][VALUE_COL].dropna()
group2 = df[df[GROUP_COL] == g2_name][VALUE_COL].dropna()

t, p = stats.ttest_ind(group1, group2)

print(g1_name, "mean:", group1.mean())
print(g2_name, "mean:", group2.mean())
print("t-statistic:", t)
print("p-value      :", p)

if p < 0.05:
    print("Result: The two group means differ significantly.")
else:
    print("Result: No significant difference between the groups.")
```

```
male mean: 30.72664459161148
female mean: 27.915708812260537
t-statistic: 2.499206354920835
p-value      : 0.012671296797013709
Result: The two group means differ significantly.
```

```
In [3]: # Program 10: Independent two-sample t-test
import pandas as pd
from scipy import stats

FILE = r"D:\Datascience-MachineLearning\Datasets\titanic.csv" # <-- change to your file path
GROUP_COL = "Survived" # column that splits the data in two groups
VALUE_COL = "Pclass" # numeric column to compare
```

```

df = pd.read_csv(FILE)
g1_name, g2_name = df[GROUP_COL].dropna().unique()[2]

group1 = df[df[GROUP_COL] == g1_name][VALUE_COL].dropna()
group2 = df[df[GROUP_COL] == g2_name][VALUE_COL].dropna()

t, p = stats.ttest_ind(group1, group2)

print(g1_name, "mean:", group1.mean())
print(g2_name, "mean:", group2.mean())
print("t-statistic:", t)
print("p-value    :", p)

if p < 0.05:
    print("Result: The two group means differ significantly.")
else:
    print("Result: No significant difference between the groups.")

```

```

0 mean: 2.5318761384335153
1 mean: 1.9502923976608186
t-statistic: 10.725263268949469
p-value      : 2.5370473879814566e-25
Result: The two group means differ significantly.

```

In []: